

DISTRIBUTIONS WITH COMPLETELY MONOTONE PROBABILITY SEQUENCES

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Abstract

Mixtures of geometric distributions are characterized by the property of having completely monotone probability sequences. Construction of such distributions and identifying the mixing distribution are considered. It is shown that these distributions are log-convex, a property which is stronger than infinite divisibility, and that they are closed under weak convergence. Characterizing geometric distributions using convolution equations, we prove that a mixed binomial process is renewal if and only if it is binomial.

1. Introduction

Pillai and Sandhya (1990) showed that distribution functions (d.f) with completely monotone derivative is a characteristic property of mixtures of exponentials and that they form a proper subclass of the class of geometrically infinitely divisible (GID) distributions. Also we know that every GID distribution is infinitely divisible (ID), see Sandhya (1991). Surveying ID distributions, Steutel (1973) observed that completely monotone (CM) densities are log-convex and they are ID. Further from Sandhya and Satheesh (1997), a mixed Poisson process is renewal if and only if it is Poisson, following a characterization of the exponential model using a convolution equation.

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In the discrete case, following Steutel (1973), for distributions on non-negative integers, log-convexity implies that they are compound geometric and they are ID. From Klebanov, et. al (1984) it is clear that compound geometric distributions are GID and that log-convexity implies that the distribution has decreasing hazard rate (DHR) (Johnson, et. al, 1992, p. 209).

In the present paper we discuss the discrete analogue of some properties of mixtures of exponentials. In Section 2 we prove that the only distributions with completely monotone probability sequences (CMPS) are mixtures of geometric distributions. Identification of the mixing distribution and construction of such mixtures are also discussed. In Section 3 the divisibility properties of these distributions are dealt with, where we show that CMPS's are log-convex, implying that they are GID and hence ID. In this way construction of a variety of distributions with DHR becomes easy. Through a characterization of the geometric law using convolution equations it is shown that a mixed binomial process is renewal if and only if it is binomial. We denote by $\mathcal{G}$ the class of distributions with CMPS's.

Possible applications of mixtures of discrete distributions and mixed binomial processes are discussed in Johnson, et. al (1992) and Feller (1968, 1966). The utility of the distributions discussed here are enhanced by the fact that they have DHR.

2. Mixtures of Geometric Distributions

Let Z be a geometric r.v with probability sequence (PS)

$$P(Z = k) = pq^k, \quad k = 0, 1, 2, \dots, \quad 0 < p < 1, \quad p + q = 1.$$

Then we have

$$P(Z < k) = 1 - P(Z \geq k) = 1 - q^k.$$

Randomizing q with a distribution G concentrated on $(0, 1)$, we have the d.f F of the mixture as

$$F(k) = P(X < k) = 1 - \int_0^1 q^k dG(q).$$

$$F(k) = 1 - m(k), \quad k = 0, 1, 2, \dots \quad (2.1)$$

where $\{m(k)\}$, the moment sequence of G is CM by Hausdorff's theorem. If $\{f(k)\}$ denote the PS corresponding to this mixture, then,

$$\Delta F(k) = f(k) = -\Delta m(k), \quad k = 0, 1, 2, \dots$$

and $\{f(k)\}$ is CM, Δ being the differencing operator.

Conversely, starting with a CMPS $\{f(k)\}$, the survival sequence $m(k) = P(X \geq k)$ is again CM with $m(0) = 1$. Now by Hausdorff's theorem and retracing the steps we see that $\{f(k)\}$ corresponds to a mixture of geometric distributions and we have:

Theorem 2.1. A PS $\{f(k)\}$, $k = 0, 1, 2, \dots$ is CM if and only if it is a mixture of geometric distributions.

If Y is the r.v with d.f G , the r.v $Z = -\log Y$ is distributed on $(0, \infty)$ and its d. f H is related to G by, $G(y) = 1 - H(z)$. Since

$$m(k) = \int_0^1 y^k dG(y) = \int_0^\infty e^{-kz} dH(z),$$

the Laplace transform (LT) of H evaluated at non-negative integer values of k is equal to the k^{th} moment of G . Thus the representation $F(k) = 1 - m(k)$, enables one to find $\{m(k)\}$, which viewed as a LT gives the d.f H which in turn determines G and the mixing distribution is identified uniquely.

Clearly, if $m(k)$, $k \geq 0$, is a LT, then $\{m(k)\}$, the sequence, is also CM and $m(0) = 1$. Thus we can construct a variety of mixtures of geometric distributions using (2. 1). Some illustrative cases are:

Survival sequence $\{m(k)\}$	Mixing distribution G
$\theta/(\theta + k)$, $\theta > 0$	Power function, $\theta y^{\theta-1}$
$\frac{1}{\sqrt{\pi}} \Gamma(1/2 + k)/\Gamma(1 + k)$	Arc sine
$\frac{\Gamma(p+q) \Gamma(p+k)}{\Gamma(p) \Gamma(p+q+k)}$	Beta (p, q)
$1/(1+k)^\alpha$, $\alpha > 0$	$(-\log y)^{\alpha-1}/\Gamma(\alpha)$
$\exp\{-k^\alpha\}$, $0 < \alpha \leq 1$	Distribution of $Y = \exp\{-X\}$, X being positive stable
$1/(1+k^\alpha)$, $0 < \alpha \leq 1$	Distribution of $Y = \exp\{-X\}$, X being Mittag-Leffler

Also, corresponding to each mixture of exponentials we have a mixture of geometric laws and vice-versa. Thus, for example, we can have discrete analogs of Weibull and Pareto laws and their generalizations discussed in Sandhya and Satheesh (1997).

3. Divisibility Properties

We know (Feller, 1968, p. 289) that in the case of integral valued r.v.'s compound Poisson distributions alone qualifies to be ID. Thus from Klebanov, et. al (1984) only compound geometric distributions can be GID among integral valued r.v.'s. Since a Poisson distributions has only Poisson factors (Raikov's theorem) and it is not compound geometric, Poisson distribution is not GID.

Now if $\{f(k)\}$ is a CMPS, then by Hausdorff's theorem, $f(k) = \int_0^1 x^k dM(x)$

for a finite measure M and by Schwarz' inequality for arbitrary measures

$$\int_0^1 x^{k+1} dM(x) \int_0^1 x^{k-1} dM(x) \geq \left[\int_0^1 x^k dM(x) \right]^2.$$

i.e. $f(k+1)f(k-1) \geq f^2(k)$ and hence:

Theorem 3.1. *Distributions in $\mathcal{G}$ are log-convex.*

Now, we are in a position to construct a variety of discrete distributions that are log-convex, GID, and ID with DHR. The proofs of the following are immediate.

Using (2.1) and invoking the convergence of moments (Feller, 1966, p.244) we have:

Theorem 3.2. *$\mathcal{G}$ is closed under weak convergence.*

Theorem 3.3. *A sequence $\{F_n\}$ in $\mathcal{G}$ converges to a d.f. F in $\mathcal{G}$, if and only if the sequence of mixing distributions corresponding to $\{F_n\}$ converges to the mixing distribution of F .*

Remark 3.1. Clearly the distribution of the first order statistic in a sample from F in $\mathcal{G}$, is in $\mathcal{G}$.

4. Geometric Distributions and Binomial Processes

In this section the survival sequence is taken as,

$$s(k) = f(k+1) + f(k+2) + \dots$$

and we have:

Theorem 4.1. A PS $\{f(k)\}$ satisfies the convolution equation $\{f(k)\} * \{s(k)\} = \{k f(k)\}$, $k = 0, 1, 2, \dots$ if and only if $f(k) = pq^{k-1}$, $k = 1, 2, 3, \dots$

Proof. Taking the corresponding p.g.f.'s we have

$$P(s) [1 - P(s)] = (1-s) s P'(s), \text{ (Feller, 1968, p. 256)}$$

where $P(s)$ is the p.g.f. of $\{f(k)\}$. The solution of this differential equation for a p.g.f. is:

$$P(s) = as / (a + b - bs) = ps / (1 - qs), \quad p = a / (a + b), \quad q = 1 - p.$$

Theorem 4.2. The geometric distribution with mean q/p is characterized by

$$\{f(k)\} * \{s(k)\} = \{(k+1)f(k+1)\}, \quad k = 0, 1, 2, \dots$$

Proof. Follows now, by solving $P(s) [1 - P(s)] = (1-s) P'(s)$ for the p.g.f. P .

Denoting by N_n the number of successes in the first n Bernoulli trials with success probability $p = 1 - q > 0$, we have a binomial process $\{N_n, n = 1, 2, 3, \dots\}$. Let $T^{(k)}$ stand for the number of trials up to and including the k^{th} occurrence of a success and $f(n)$, the probability that a success occurs for the first time at the n^{th} trial. Thus we have a renewal process with waiting time distribution $\{f(n), n = 0, 1, 2, \dots\}$, $f(0) = 0$, and the identity,

$$\begin{aligned} P\{N_n \geq k\} &= P\{T^{(k)} \leq n\} \\ &= F^{k*}(n), \quad k = 0, 1, 2, \dots, n \quad \text{and} \quad n = 0, 1, 2, \dots \end{aligned}$$

with $F^{1*}(n) = F(n) = f(0) + f(1) + \dots + f(n)$.

Now, the content of example (a) in Feller, 1968, p. 313 is:

Theorem 4.3 *A counting process is binomial if and only if the waiting times are geometric.*

Now randomizing the parameter q in the binomial process with a distribution G concentrated on $(0, 1)$ we have a mixed binomial process $\{N_n\}$ and

$$\begin{aligned} P\{N_n = k\} &= \binom{n}{k} \int_0^1 q^{n-k} (1-q)^k dG(q) \\ &= \binom{n}{k} (-1)^k \Delta^k m(n-k) \end{aligned}$$

where $m(\cdot)$ is the moment sequence of G . Now, if $\{N_n\}$ is renewal with waiting time d.f. F , then

$$P\{N_n = k\} = \binom{n}{k} (-1)^k \Delta^k m(n-k) = F^{k*}(n) - F^{(k+1)*}(n).$$

for

$$k = 0, P\{N_n = 0\} = m(n) = 1 - F(n) \quad (4.1)$$

for

$$k = 1, P\{N_n = 1\} = -n \Delta m(n-1) = F(n) - F^{2*}(n).$$

Using (4.1)

$$n \Delta F(n-1) = F(n) * [1 - F(n)]$$

or,

$$\{f(n)\} * \{s(n)\} = \{nf(n)\}.$$

Now, invoking Theorems 4.1 and 4.3 we have,

Theorem 4.4. *A mixed binomial process is renewal if and only if it is binomial.*

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